

The examination is being carried out on the **following application documents**

Description, Pages

1-27 as originally filed

Claims, Numbers

1-23 as originally filed

Drawings, Sheets

1/4-4/4 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

- D1 DE 199 34 169 A1 (HENKEL KGAA [DE]) 25 January 2001 (2001-01-25)
- D2 ROY CHOUDHURY SAMRAT ET AL: "Investigation of antimicrobial physiology of orthorhombic and monoclinic nanoallotropes of sulfur at the interface of transcriptome and metabolome",
APPLIED MICROBIOLOGY AND BIOTECHNOLOGY, SPRINGER, DE,
vol. 97, no. 13, 2 April 2013 (2013-04-02), pages 5965-5978,
XP035328837,
ISSN: 0175-7598, DOI: 10.1007/S00253-013-4789-X
[retrieved on 2013-04-02]
- D3 US 2 460 776 A (VINCENT JAMES G) 1 February 1949 (1949-02-01)
- D4 US 4 192 866 A (ANDERSON RALPH [US]) 11 March 1980
(1980-03-11)
- D5 WO 2011/044568 A1 (WILLIAM MIZELL LIGON [US]) 14 April 2011
(2011-04-14)
- D6 DATABASE TKDL [Online] 15 December 2009 (2009-12-15), nn: "Sudha Parpati",
XP002769053, Database accession no. AK15/113

- D7 DATABASE TKDL [Online] 15 June 2012 (2012-06-15), nn: "Vajra^oekhararasa^a (Prathama^a)", XP002769054, Database accession no. VS/1093B
- D8 DATABASE TKDL [Online] 15 December 2009 (2009-12-15), nn: "Ma^oji⁴h^odyamtailam (3)", XP002769055, Database accession no. RS/295
- D9 DATABASE TKDL [Online] 15 December 2009 (2009-12-15), nn: "Mamsi Taila", XP002769056, Database accession no. AK11/2851
- D10 EP 0 281 812 A1 (MILOR SCIENT LTD [US]) 14 September 1988 (1988-09-14)
- D11 COLE H N: "TOPICAL THERAPY", CUTIS, EXCERPTA MEDICA, BELLE MEAD,NJ, US, vol. 14, no. 3, 1 January 1974 (1974-01-01), pages 453-455, XP002044491, ISSN: 0011-4162
- D12 WO 02/02059 A1 (MEDICIS PHARMACEUTICALS [US]) 10 January 2002 (2002-01-10)

1 Preliminary remarks:

1.1 The term "polymeric [sulfur Sn]" used in claim 1 (and also in claims 2-4) is vague and unclear and leaves the reader in doubt as to the meaning of the technical feature to which it refers, thereby rendering the definition of the subject-matter of said claim unclear (Article 84 EPC).

The description p.3 further mentions "long chains of sulfur atoms" (p.15) and solubility criteria (l.20ff "insoluble") which are not more clear than the term "polymeric"/"polymerized", since no minimum number of sulfur atoms is given (apart from the limitations defined in present dependent claims 6 or 7), nor are solubilizing parameters (solvent, temperature etc.) specified (this applies also for the definition of "soluble sulfur" in claim 17).

The commercial product "Polymeric Sulfur (Sn) IS90 from Wuxi Wanli Chemical, China" as mentioned within the context of the example section (p. 18ff) is appearing to be a registered trade mark that has no precise meaning

as it is not internationally accepted as a standard descriptive term, and specifications about (average) molecular weight or "partial solubility in solvents other than water" are not available.

The application further mentions (p.2, l.29-31) that "Sulfur exists at room temperatures primarily as rhombic crystals. Other forms of sulfur, such as monoclinic crystalline sulfur, or polymeric sulfur, are the normal primary forms which elemental sulfur assumes at certain higher temperature ranges. At room temperatures, these forms convert, or revert, to rhombic sulfur."

As a consequence, the available prior art was assessed in view of processing sulfur and/or compositions comprising it "at certain higher temperature ranges", which invariably lead to sulfur and/or compositions comprising certain amounts of "polymeric sulfur" as claimed.

1.2 The subject matter of claims 19-23 concern methods of treatment of the human or animal body by surgery or therapy or to a diagnostic method practised on the human or animal body, which is excluded from patentability (Article 53(c) EPC). The claims as presently formulated are thus not allowable and should be either suppressed or reworded in the form of a product for use in such a method according to Articles 54(4) and 54(5) EPC (cf. Guidelines G-II, 4.2 and G-VI.7).

1.3 The term "substantially free of [non-cosmetic oils etc.] employed in claims 2-4 is vague and unclear and leaves the reader in doubt as to the meaning of the technical feature to which it refers, thereby rendering the definition of the subject-matter of said claim unclear (Article 84 EPC).

Only the definition given on p.9, l.14-17, which specifies weight% ranges, would be a clear definition of the respective technical feature, which until then is interpreted in the broadest possible way.

2 Novelty: Art.52(1), 54 EPC

2.1 Claim 1 is defines

"A personal care composition comprising polymeric sulfur Sn."

2.2 The subject-matter of claim 1 cannot be delimited from (see item 1.1 above) and is therefore not novel over the (at least implicit) disclosure of D1-D5:

- 2.2.1 D1 (claims) discloses cosmetic and/or dermatological compositions comprising nanoparticulate sulfur (10-1500 nm) with enhanced stability and efficacy (col.1, l.17-25 and l.46-57). The nanoparticulate sulfur is prepared e.g. by reacting H₂S and SO₂ (claim 8) which has been indicated in the description of the present application as a way to obtain "amorphous"/"insoluble" sulfur (p.3, l.23-25: "Insoluble sulfur is an amorphous form of sulfur, which is made from the heat-polymerization of sulfur and also can be obtained by reacting hydrogen sulfide with sulfur dioxide").
- Example 3 (col.7) describes a "salve for treating seborrhea and acne vulgaris" comprising said nanoparticulate sulfur which is implicitly "polymeric" as claimed.
- 2.2.2 D2 (abstract; introduction) claims to elucidate mechanisms responsible for the antimicrobial effects (tab.1) of orthorhombic (i.e. soluble; "alpha-SNP") and monoclinic (i.e. polymeric/insoluble; "beta-SNP") nanoallotropes of sulfur (see Fig.2).
- The beta-SNP's were synthesized via W/O microemulsion technique and encapsulated within the conjugate of Span-80 and Tween-80 nonionic surfactants, which are suitable as personal care compositions.
- 2.2.3 D3 (claim 6 and other claims) relates to fluid compositions comprising "fused bentonite sulfur compositions" (col.1, l.36-45) having active bacteriostatic, bactericidal and fungicidal properties and being suitable for treatment of various skin infections e.g. acne (col.8, l.68 - col.9, l.4).
- A topical cream formulation is illustrated on col.7, l.45-57.
- 2.2.4 D4 (claims 2,3) discloses a topical composition (col.1, l.9-11) for treating i.a. inflammation, which comprises a fatty basis (animal or vegetable) and precipitated sulfur (col.2, l.57-64); the preparation (col.3, l.53-55 and examples) involves heating the sulfur powder sprinkled on top of the heated fatty base composition before stirring it into the bulk of the composition. This implies the presence of certain quantities of "polymeric" sulfur (see item 1.1) in the resulting compositions.
- 2.2.5 D5 (claims 4 and 7) disclosed a topical skin cream comprising a mixture of sublimed sulfur powder, petroleum jelly and mineral oil for treating a variety of skin conditions incl. tinea versicolor, eczema etc. (p.2, par.4; p.5, par.14 - p.6,

par.20). The preparation method (claims 4 and 7) includes at least one heating step and thus implies the presence of certain quantities of "polymeric" sulfur (see item 1.1) in the resulting compositions.

2.3 In addition, D6-D9 prove that "polymeric sulfur Sn" as it is construable (see item 1.1 above) has indeed been known "for many years" and is in fact part of Traditional Knowledge of the Indian subcontinent for treating acne and other dermatological conditions.

2.4 The gist of the method of treating topical skin conditions as defined in claim 19 and claims 20-23 dependent thereof (see however item 1.2) is also not novel over D1-D9.

2.5 It is noted that limitations as defined in dependent claims 9-18 are also foreseen in D1-D9 or else, suggested for the same purpose in D10-D12.

3 Inventive step: Art.52(1), Art.56 EPC

Although the presence of an inventive step can be justified only when novelty has been unambiguously established, the Applicant is presented with the following preliminary analysis:

3.1 The present application puts forward as problem(s) to be solved to provide sulfur compositions having reduced irritation and enhanced bacterial properties (p.3, l.10-11).

3.2 The example section (p.18) provides data for antimicrobial efficacy tests, anti-inflammatory potential, mildness for a single type of "polymeric sulfur" (see item 1.1 above) which has as yet not been characterized, and in comparison with so-called "regular sulfur" (ex Sigma; no specs given) and a commercial product called "Lonza KGM-Gold Bullet Kit" (no specs given apart from the statement that it is "without hydrocortisone").

Although some effects seem to be discernible for this set-up (see figures), it is noted that

- a single type of "polymeric sulfur" has been used, raising doubts as to whether the problem is solved over the whole scope as claimed, e.g. also for "sulfur Sn with n at least 9" as defined in present dependent claim 6;

- no comparison with the prior art cited herein has as yet been provided, notably not with sulfur treated the traditional way according to D6-D9.

In the absence of any surprising technical effect over the existing prior art (any of D1-D9), the objective technical problem will remain the provision of alternative sulfur compositions for topical application.

At least D1 and/or D2 would enable the skilled person to arrive at the anti-microbial topical compositions of the present application.

For any limitation according to dependent claims (e.g. 9-18), it would also be necessary to demonstrate surprising technical effects in order to make it plausible that such technical features are not obvious in view of D1-D9 and/or D10-D12.

The examination is being carried out on the **following application documents**

Description, Pages

2, 4, 7, 10-27 as originally filed

1, 3, 5, 6, 8, 9 filed in electronic form on 20-03-2018

Claims, Numbers

1-19 filed in electronic form on 20-03-2018

Drawings, Sheets

1/4-4/4 as originally filed

1 Art.123(2) EPC

The amendments realised by the Applicant are based on the application as filed, and do not lead to extension of the subject-matter.

However, the amendments in claims 2-4 are incomplete, since "(contains less than 1%) by weight (of ...)" is missing, although the passage indicated by the applicant as basis for these amendments (p.9, l.14ff) do contain this wording.

2 Art.84 EPC

2.1 Present claim 8 (identical to original claim 12) is unclear because of the arbitrary punctuation, which does not allow to discern the actual names of the different cysteine derivatives listed. In addition - or because of that - "L-Cysteine" is listed several times in an apparently redundant way.

2.2 Unclear wording such as "about" within the context of amounts, or "and the like" within the context of lists should be removed from the application.

3 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 does not involve an inventive step within the meaning of Article 56 EPC:

3.1 Closest prior art

The present application puts forward as problem(s) to be solved to provide sulfur compositions having reduced irritation and enhanced bacterial properties (p.3, l.10-11).

D1-D3 disclose the antimicrobial properties of polymeric sulfur; D1 (examples) and D3 (col.8, l.74) disclose the use in the treatment of acne.

D1 mentions problem of incompatibility of elemental sulfur with common cosmetic and dermatological formulation components (col.1, l.17-25 and l. 46-56), and the aim of providing compositions with higher cosmetic and/or dermatological efficacy, which at least implies tolerance of the treated skin towards actives. Examples 2 and 3 of D1 comprise nanoparticulate sulfur prepared in example 1 ("sulfur gel" = amorphous, hence polymeric sulfur, see col.7, l.8).

3.2 Technical effect and problem to be solved

The subject-matter of claim 1 differs from examples 2 and 3 of D1 in that the composition further comprises a cysteine.

3.2.1 The corrected example 2 as filed with letter dated 20.03.2018 shows that polymeric sulfur Sn at 1% and complemented with 25mM L-Cysteine was more effective against P.acnes than the same amount of regular sulfur S with 25mM L-Cysteine.

However, the result for 0,1%Sn and 25mM L-Cysteine in said corrected example 2 does not differ significantly from the result for 0,1% regular sulfur 25mM L-Cysteine. It is concluded that there seems to be an effective minimum concentration for polymeric sulfur.

The examining division is therefore of the opinion that the problem of enhanced bacterial properties as stated in the application is in fact not solved over the whole scope of present claim 1.

3.2.2 The examples 8 and 9, and figures 1-4 mentioned by the applicant in the letter dated 20.03.2018 as showing reduced irritation also seem to provide comparison with "regular" sulfur.

3.2.3 However, this means that Examples 2, 8 or 9 do not constitute a valid comparison with the closest prior art represented the examples of D1, where polymeric sulfur is already present, and also used with further compounds/ actives in the topical cream formulations described therein.

The effects illustrated by examples 2, 8 or 9 therefore cannot be taken into account when formulating the objective technical problem, which is rather the provision of alternative (anti)bacterial topical compositions based on polymeric sulfur.

3.2.4 The addition of cysteine as further compound to compositions for treating topical skin conditions comprising polymeric sulfur is considered to be arbitrary, not the least, because the traditional remedies mentioned in D6-D9 mention milk(products) as additives and/or processing aids when preparing the final acne remedy, which implies the presence of cysteine from the animal protein present in such compositions.

Moreover, D10 explicitly teaches that the amino acid cysteine is a suitable anti-inflammatory agent to be included into compositions for the treatment of acne, for "mild anti-inflammatory" action (p.4, col.5, l.30ff), with cysteine even being preferred (p.4, col.5, l.53-54).

When looking for alternatives, the skilled person does not need more incentive for considering cysteine as a further component in compositions for treating topical skin conditions comprising polymeric sulfur.

The applicant is given another opportunity to react in writing to the objections put forward above.

Should the applicant insist on the present set of claims, or not remedy all objections raised during the procedure, summons for oral proceedings shall be issued in accordance with the request filed with 20.03.2018 (Art.116 EPC).